

UNITED INTERNATIONAL UNIVERSITY

CSE 2213 (DISCRETE MATHEMATICS) – CLASS TEST 1

TIME: 30 MINUTES, FULL MARKS: 20

There are 2 questions. **Answer all questions.** Figures located in the right-hand margin indicate full marks.

1. Using mathematical induction, prove that for all positive values of n , 10
$$1 \times 1! + 2 \times 2! + \dots + n \times n! = (n + 1)! - 1$$

2. Using mathematical induction, prove that $n^3 - n$ is divisible by 6 whenever n is a positive integer. 10

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CSE 2213 (DISCRETE MATHEMATICS) – CLASS TEST 1

TIME: 40 MINUTES, FULL MARKS: 40

There are 2 questions. **Answer all questions.** Figures located in the right-hand margin indicate full marks.

1. Using mathematical induction, prove that for all positive values of n , 10

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$$

2. Using mathematical induction, prove that $3^n - 1$ is divisible by 2 whenever n is a positive integer. 10